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Dam defects ontology and knowledge graph construction from Multi-Source hazard data using large language models

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ABSTRACT

During dam operation, large volumes of information on dam defects are documented in various paper-based reports. This information is essential for effective dam safety management, yet its dispersed distribution and unstructured textual form make it difficult to transform into structured and reusable knowledge. Therefore, it is imperative to mine knowledge from these resources to construct a knowledge graph for effective utilization. This study presents the construction and application of such a graph by leveraging large language models (LLMs). A dam defects ontology (DDO) was designed to systematically model dam entity properties, spatial distribution, technical parameters, and safety hazards, with particular emphasis on detailed hierarchical knowledge modeling of engineering hazard forms and their components. To mitigate the "hallucination" problem of LLMs in the specialized domain of dam safety, a domain-adaptive Ontology-guided Hazard Classification Framework (OHCF) was developed. OHCF integrates four core modules—fine-tuning, temperature-perturbed voting, ontology-constraint consistency checking, and expert validation—to improve the accuracy of dam hazard entity recognition. In addition, OHCF dynamically expands the DDO through a semi-supervised mechanism. Experimental results show that all evaluation metrics of OHCF exceed 90 %. Based on the proposed method, this study mined dam safety evaluation reports for 22,076 defective and dangerous dams in China and constructed a dam defects knowledge graph (DDKG). Moreover, the proposed DDKG exhibits broad application potential in dam hazard risk analysis and in supporting expert decision-making for dam safety management.

1. Introduction

Reservoir and dam projects are globally important infrastructure, providing multiple functions such as flood control and disaster mitigation, water supply, agricultural irrigation, and hydropower generation. The safety of these projects is directly related to social stability. With the increase in operating years, dam projects commonly experience structural aging and functional degradation [1]. Coupled with the current frequent occurrence of natural disasters and intensified human activities, the safety situation of dams is continuously becoming more severe. Traditional risk analysis methods are mostly oriented toward individual projects and rely heavily on expert judgment or numerical computation [2–4]. Analyses of risk factors across multiple projects at regional scales remain limited. The emergence of data-driven methods has introduced new perspectives for risk assessment [5–7]. Using advanced knowledge engineering techniques, extensive amounts of dispersed hazard information can be structurally reorganized, enabling the identification of deep correlations. This provides a solid knowledge foundation for risk tracing and scientific decision-making.

At present, much of the critical key information related to dam defects is recorded in unstructured text [8]. These textual data are heterogeneous in format and poorly structured, making it difficult to effectively apply the embedded knowledge to correlation analysis and reasoning. This has led to the dilemma of being "rich in data but poor in knowledge," which constrains in-depth and systematic knowledge discovery of the mechanisms of dam defects and hazards.

Knowledge graphs (KGs) [9] offer an efficient tool for transforming this information into structured knowledge for organization and

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analysis. They can reveal complex semantic relationships among entities and provide a feasible solution to the above problem. Constructing KGs from large volumes of engineering text relies on effective knowledge extraction. In this regard, deep learning-based natural language processing has made significant progress [10]. In particular, recent advances in large language models (LLMs) have created new opportunities for large-scale knowledge extraction from massive text corpora [11,12]. LLMs exhibit strong capabilities in contextual understanding and reasoning, and can perform tasks by following natural language instructions[13–16].

However, when applied to the specialized domain of dam engineering, LLMs face combined challenges related to domain adaptability, knowledge complexity, and application economics [17]. General-purpose models lack a deep understanding of specialized dam engineering knowledge, which hinders accurate mapping of domain-specific terminology and parsing complex hierarchical relationships, often resulting in factual errors. Moreover, applying large-scale models for dataset processing incurs high computational costs, limiting their efficiency and scalability in practical engineering applications.

To address these challenges, this study develops a dam defects ontology (DDO) to provide a structured representation framework for dam hazard knowledge. An Ontology-guided Hazard Classification Framework (OHCF) is further proposed, which leverages LLMs to standardize the classification of unstructured hazard descriptions. Based on this process, a dam defects knowledge graph (DDKG) is constructed to achieve the structured integration of hazard information across multiple dam projects, thereby supporting dam safety management.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the proposed methodology. Section 4 evaluates model performance and presents case studies using the KG. Section 5 concludes the study and outlines future research directions.

2. Related works

2.1. Knowledge representation in hydraulic engineering

With the deepening application of digital twin technology in the hydraulic sector, the efficient organization and utilization of massive information throughout the entire engineering lifecycle have become a pivotal challenge for the industry's intelligent advancement. Consequently, numerous studies have focused on developing specialized ontologies to represent relevant data and requirements in a structured manner.

As an explicit and formal specification of a shared conceptualization [18], an ontology serves as a fundamental pillar of the Semantic Web infrastructure and plays a critical role in the construction of knowledge graphs [19]. In the engineering domain, ontologies are introduced to model specialized knowledge and provide a robust organizational framework for domain-specific data. Given the inherent complexity of ontology construction, evaluating and reusing existing domain ontologies has become a crucial preliminary step for researchers embarking on new projects.

To date, various ontologies tailored for hydraulic engineering safety have been developed. Zhang et al. [20] constructed a knowledge ontology for hydraulic engineering safety management centered on the causes of engineering hazards and corresponding countermeasures, providing a structured knowledge model to enhance decision-making in risk management. Gan et al. [21] developed a multi-layered, ontology-driven KG framework for the resilience enhancement of underground structures, integrating resilience quantification and disaster chain analysis to bridge the gap between subjective experience and intelligent decision-making. Utilizing unstructured data—including risk prevention manuals and design schemes from the Middle Route of the South-to-North Water Diversion Project—Liu et al. [22] developed an emergency plan ontology, which provides key support for the intelligent generation and reuse of emergency strategies. Focusing on the four dimensions of "structure-process-environment-material," Zhou et al. [23] proposed a unified knowledge management framework by bridging Building Information Modeling and domain ontologies to address the fragmentation of heterogeneous monitoring data, enabling efficient knowledge extraction and semantic integration in dam safety systems. Furthermore, Baudrit et al. [24] developed a holistic, graph-based knowledge model for dam systems by utilizing the formalism of conceptual graphs to unify heterogeneous, cross-disciplinary knowledge flux, providing a convenient framework for capitalizing experience, predicting structural states, and supporting forensic civil engineering.

While the aforementioned studies all aim to enhance the operation and maintenance of hydraulic engineering, they vary significantly in their representation granularity and thematic focus. In general, existing research still falls short in characterizing the specific manifestations of defects and the intricate relationships between relevant influencing factors. This limitation hinders the capacity of current models to support fine-grained analysis and advanced reasoning within the domain of dam safety.

2.2. Construction of knowledge graphs

In 2012, Google introduced the concept of the KGs [25], which has since been extensively researched and applied across various domains [26–29]. KG is a structured semantic network that depicts real-world knowledge through "entity-relation-entity" triplets. While an ontology serves as the conceptual framework defining the schema layer of knowledge, KG represents the knowledge base instantiated by real-world data within this framework.

A key step in constructing the KG data layer is the extraction of domain entities from unstructured text, which is commonly formulated as a Named Entity Recognition (NER) task in Natural Language Processing (NLP). NER aims to identify entity mentions in text and assign them predefined entity type labels. Early NER methods typically relied on complex manually crafted rules [30–32]. With the rise of deep learning, NER systems achieved substantial performance improvements by automatically learning textual feature representations. Recurrent Neural Networks [33,34] and Convolutional Neural Networks [35–37] were initially used to encode contextual information. Subsequently, the Transformer [38] architecture, built on self-attention mechanisms, led to the development of Pre-trained Language Models (PLMs) such as BERT [39]. Through supervised fine-tuning on labeled datasets, PLM-based methods have demonstrated strong performance in various NER tasks [40–42]. However, such methods usually depend on substantial high-quality annotated data and well-defined label systems [43]. When these label systems are constrained by limited expert knowledge, the resulting models may struggle to generalize to domain-specific concepts that are weakly represented, ambiguously defined, or absent from the predefined schema.

Recently, the application of LLMs in NER has gained widespread attention [44]. Having undergone deep pre-training on massive and diverse unlabeled data, LLMs possess robust generalization capabilities and extensive linguistic knowledge. In downstream tasks like NER, LLMs can achieve superior performance guided solely by natural language instructions. Numerous studies have utilized LLMs to assist in KG construction [45–49]. Mainstream methodologies typically rely on meticulously designed structured prompts, combining few-shot examples with the target text to guide the model in generating standardized outputs for extracting entities, attributes, and relationships. Furthermore, research has demonstrated that LLMs can effectively assist in the construction and expansion of ontology models [50,51]. Nevertheless, the direct application of LLMs in specialized domains such as hydraulic engineering remains challenging. This is primarily due to the distribution shift between the training data and domain-specific contexts, resulting in insufficient mastery of specialized terminology and domain knowledge [52]. To enhance the utility of models in specific fields, recent studies have introduced strategies such as Parameter-Efficient

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Fine-Tuning [53] and Chain-of-Thought [54], effectively improving the structural regularity of model outputs and the ability to capture complex semantics.

In summary, while LLMs offer new possibilities for NER, their inherent hallucination issues may lead to the incorrect or arbitrary assignment of entity labels, thereby compromising the reliability of extraction results and hindering the structured organization and integration of knowledge.

2.3. Knowledge gap

The above studies show growing interest in building hydraulic engineering knowledge systems to support intelligent safety management and in using large language models for domain-specific knowledge processing. However, several research gaps remain. First, although existing studies in hydraulic engineering have laid a solid foundation for ontology construction, they are still insufficient for the fine-grained representation of dam defects, particularly in modeling hazard forms and affected components. Second, existing knowledge extraction methods based on PLMs, such as BERT, remain insufficiently flexible for dam engineering texts. Hazard descriptions in such texts are often fragmented, nested, and expressed in diverse ways, which increases the difficulty of knowledge extraction. Although these methods have shown good performance in classification tasks with a small number of labels, they usually require substantially more annotated data when applied to tasks involving larger label sets and more complex class relationships. In addition, these models are typically constrained by predefined categories derived from expert knowledge, making it difficult to identify previously undefined concepts. Third, although large language models have shown strong potential in NER tasks, their direct application to dam safety scenarios is still limited by insufficient domain knowledge, which often leads to significant hallucination problems.

To address these gaps, this study develops a fine-grained dam defect ontology and proposes OHCF to support ontology-guided hazard knowledge recognition and ontology expansion for dam defect knowledge graph construction.

3. Methodology

This section describes the implementation of ontology construction, data layer development, and KG storage. Fig. 1 illustrates the overall construction framework of the DDKG, which comprises three main components: DDO construction, knowledge extraction, and knowledge fusion.

3.1. DDO construction

An ontology is a standardized representation of concepts, entities, attributes, and their semantic relationships within a specific domain. In the construction of KGs, the ontology layer serves as the core structural backbone. The clarity and completeness of its design directly determine its knowledge coverage and representational accuracy of the KG [55]. In dam engineering, developing a structured and standardized ontology for defects is essential for unifying hazard terminology, enabling correlation analysis of complex causative mechanisms, and supporting downstream applications such as intelligent diagnosis and decision-making.

In this study, a seven-step methodology is adopted to construct the DDO. The process comprises the following steps: (1) defining the dam hazard domain ontology and related tasks; (2) considering the reuse of existing ontology models related to dam hazards; (3) enumerating relevant terminology; (4) defining ontology classes and hierarchical structures; (5) defining class attributes and relationships; (6) specifying constraints; and (7) creating instances.

The DDO is constructed using the Web Ontology Language (OWL) and is stored and managed with the Protégé tool. OWL is an ontology language recommended by the World Wide Web Consortium (W3C) for defining vocabularies and describing resources. By specifying classes, properties, and values, OWL provides a standardized method for users to

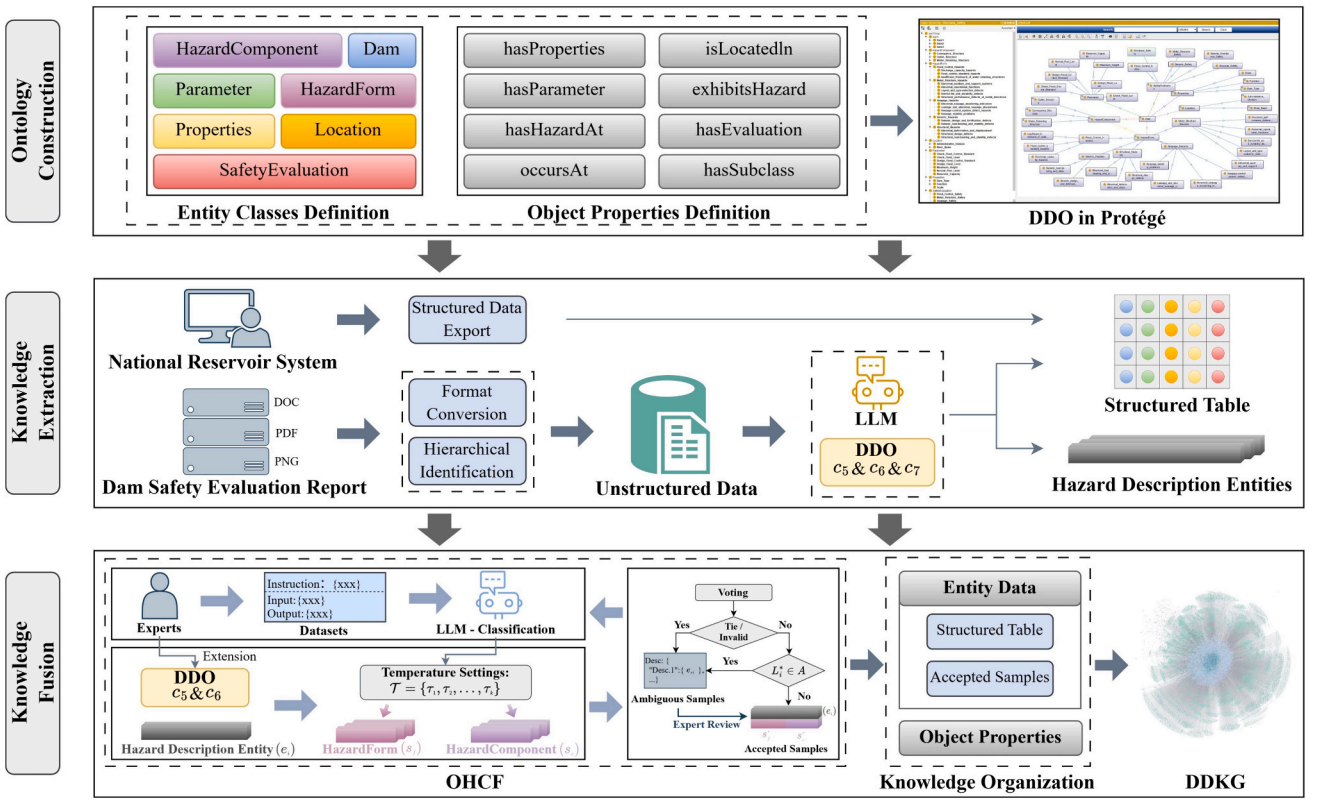


Fig. 1. Construction framework of DDKG.

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describe domain-specific classes and attributes.

The structure of the DDO can be represented using the triple form shown in Equation (1):

$$DDO = \langle C, P, H \rangle \tag{1}$$

where $C = \{c_i | i \in N^*\}$ is the set of top-level classes; c_i is the i−th top-level class; $P = \{p_{ij} | i, j \in N^*, i \neq j\}$ is the set of object properties, where p_{ij} represents a directed relationship $c_i \rightarrow c_j$; $H = \{S_i | c_i \in C^*\}$ is the set of all subclasses under each top-level class, where $S_i = \{c_i(j_1, j_2, \ldots, j_k) | i, k, j_k \in N^*\}$ represents the subclass set under top-level class c_i, and $c_i(j_1, j_2, \cdots, j_k)$ denotes a path index sequence (e.g., $c_3(2,1)$ represents the first child of the second subclass under top-level class c_3).

Dam defects arise from the coupled effects of intrinsic engineering factors and environmental conditions. Therefore, seven top-level classes are initially defined in the DDO, as summarized in Table 1.

To further enhance the semantic expressiveness and structural rigor of the DDO, both semantic and hierarchical relationships are constructed. Semantic relationships are primarily established by defining object properties to link different top-level entity classes. Table 2 provides the object properties defined in the ontology structure. All top-level classes are connected to the Dam entity to anchor all related information to specific engineering instances. Specifically, HazardForm is linked to HazardComponent via "occursAt.".

Hierarchical relationships are constructed using "hasSubclass" to form internal hierarchical structures for each top-level class, aiming to increase the depth and granularity of knowledge representation. For example, subclasses such as "dam height" and "reservoir capacity" are defined under the Parameter class. Hierarchical relationships allow subclasses to inherit object properties from their parent classes. This effectively improves the reusability, scalability, and reasoning capability of the model. Fig. 2 shows the DDO modeling in Protégé.

Among the top-level classes, "c_5 HazardForm" and "c_6 HazardComponent" are directly related to engineering hazards and constitute the most critical elements of the DDKG. These entities must be extracted from unstructured text, which poses considerable challenges. For instance, although LLMs can recognize entities such as "spillway" and "Side wall," they often fail to correctly parse the intrinsic "whole–part" logic between them. LLMs may incorrectly classify entities with subordinate relationships as being on the same hierarchical level, thereby overlooking the fact that the "Side wall" is a constituent subunit of the "spillway." Such parsing errors can directly compromise the integrity of the DDKG's knowledge structure.

To overcome these limitations in semantic understanding, this study refines the hierarchical construction of top-level classes c_5 and c_6 based on expert knowledge. The detailed construction is described below.

(a) Hierarchical structure of HazardForm.

To ensure scientific rigor and validity of hierarchical structures, it is generally needed to reference industry technical standards or national/regional administrative regulations. In China, for example, the *Guidelines on Dam Safety Evaluation* (SL 258–2017) specify that the safety of a dam project is assessed based on five categories of specialized indicators: flood control, seepage, structural integrity, seismic performance, and metal structure. These indicators provide a standardized basis for dam safety evaluation.

Table 1
Top-level entity classes of the DDO.

c_i	Entity class
c_1	Dam
c_2	Properties
c_3	Location
c_4	Parameter
c_5	HazardForm
c_6	HazardComponent
c_7	SafetyEvaluation

Table 2
Definitions of object properties in the DDO.

p_{ij}	Object properties	Domains	Ranges
p_{12}	hasProperties	c_1	c_2
p_{13}	isLocatedIn	c_1	c_3
p_{14}	hasParameter	c_1	c_4
p_{15}	exhibitsHazard	c_1	c_5
p_{16}	hasHazardAt	c_1	c_6
p_{17}	hasEvaluation	c_1	c_7
p_{56}	occursAt	c_5	c_6

As a result, hazard issues corresponding to these five categories are defined as level-1 subclasses of HazardForm. Under each level-1 subclass, level-2 and level-3 subclasses are further defined to capture detailed manifestations of engineering hazards. Table 3 presents the hierarchical structure of the HazardForm class.

(b) Hierarchical structure of Hazard Component.

The HazardComponent class is used to locate the specific structural positions where engineering hazards occur. Dam engineering structures can be broadly classified into water-retaining structures, outlet structures, and conveyance structures according to their functions. Water-retaining structures include the dam body, dam foundation, and dam abutment. Outlet structures comprise spillways, flood discharge tunnels, and bottom outlets. Conveyance structures consist of intake structures, conveyance pipelines, and control structures. Table 4 lists the hierarchical structure of the HazardComponent class in detail, with the three types of structures serving as its level-1 subclasses.

In practical applications, hierarchical modeling of complex engineering entities such as a dam project based solely on expert experience is inevitably constrained by knowledge boundaries. As a result, the ontology coverage may be limited. To address this, during knowledge extraction, the ontology is conceptually supplemented or hierarchically expanded based on identified undefined entities, as described in Section 3.2.3.

3.2. Data layer construction

The data layer is designed to extract knowledge from multi-source heterogeneous data generated during the long-term operation of reservoir dam engineering systems, based on the DDO. This section presents the construction of the DDKG data layer and provides a detailed description of OHCF. The overall framework of data layer construction is displayed in Fig. 3.

3.2.1. Knowledge sources and preprocessing

The China Reservoir Operation and Management Platform stores a large volume of fundamental engineering data in the form of structured tables, which largely satisfy the data requirements of top-level classes c_1–c_4 in the DDO.

In contrast, the data corresponding to top-level classes c_5–c_7 are derived from unstructured text. These texts are highly heterogenous in formats, including PDF, DOC, and PNG files, and the original documents often contain redundant or irrelevant information. Prior to knowledge extraction, the texts are preprocessed to eliminate noise and normalize the data.

For a given unstructured text D, its format is first converted into a structured markdown representation D_{md}. State-of-the-art optical character recognition models [56] are then applied to perform layout parsing and format conversion. Next, regular expressions are used to delimit the boundaries of text blocks $D_{md,i}$ that contain relevant knowledge entities and remove redundant content. Finally, the extracted text blocks are integrated into a single corpus, denoted as T:

$$T = \{D_{md,1}, D_{md,2}, \cdots, D_{md,n}\} \tag{2}$$

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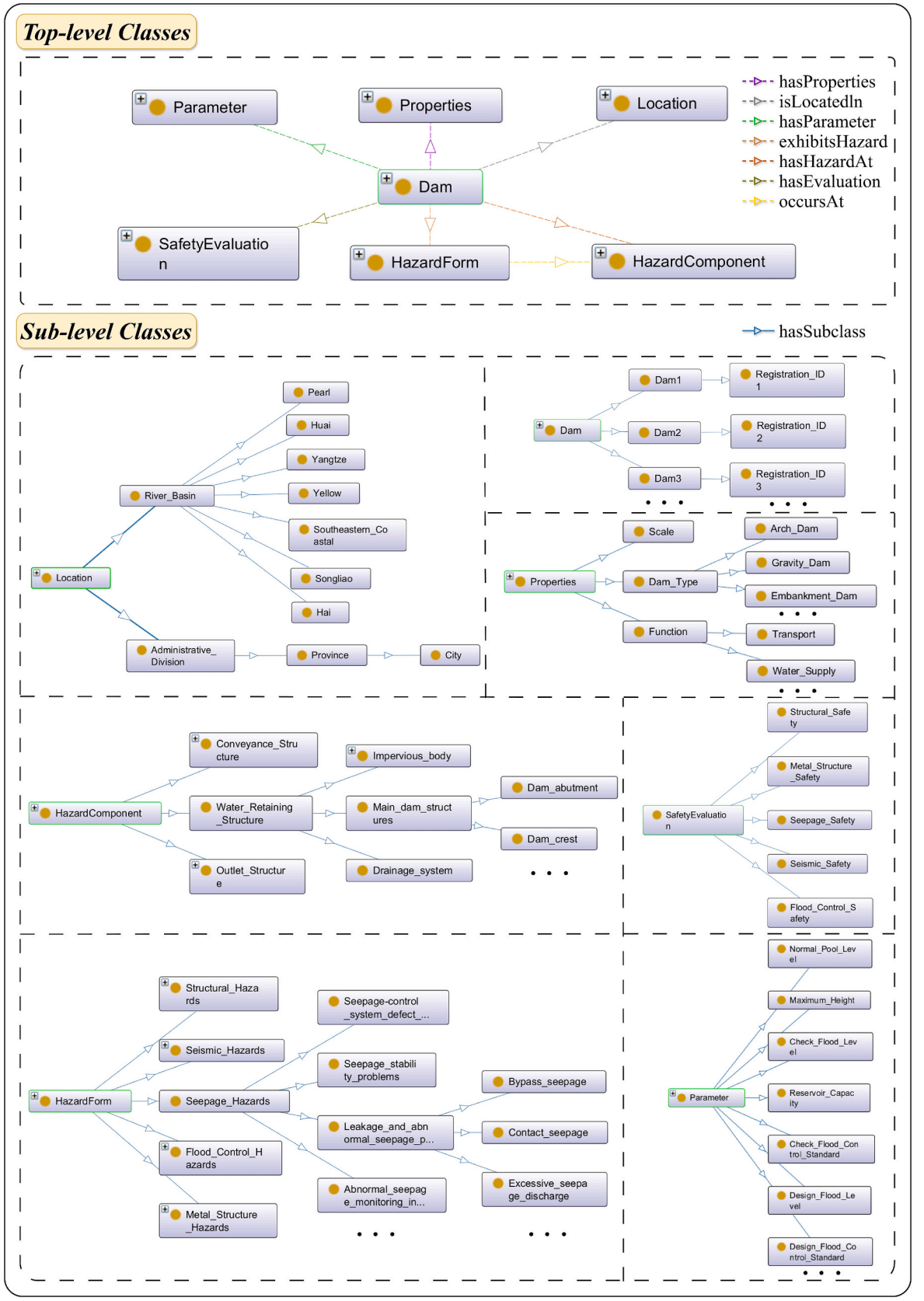


Fig. 2. DDO in Protégé.

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Table 3
Subclasses of "c_5 Hazard Form".

Subclass (level 1)	Subclass (level 2)	Subclass (level 3)
c_5(1) Flood Control Hazards	c_5(1,1) Flood control standard hazards	c_5(1,1,1) Floods exceeding design standard; c_5(1,1,2) Insufficient design flood standard; c_5(1,1,3) Design flood estimates fail to meet requirements; c_5(1,1,4) Failure to meet the requirements for recent extraordinary flood operation standards
	c_5(1,2) Insufficient freeboard of water-retaining structures	c_5(1,2,1) Insufficient water-retaining elevation; c_5(1,2,2) Incomplete enclosure of water-retaining structures
	c_5(1,3) Discharge capacity hazards	c_5(1,3,1) Missing flood control facilities; c_5(1,3,2) Insufficient discharge capacity
c_5(2) Seepage Hazards	c_5(2,1) Seepage-control system defect hazards	c_5(2,1,1) Incomplete or failed impervious body; c_5(2,1,2) Inadequate grouting; c_5(2,1,3) Abnormal filter and drainage facilities; c_5(2,1,4) Seepage-control defects at contact zones
	c_5(2,2) Leakage and abnormal seepage phenomena	c_5(2,2,1) Excessive seepage discharge; c_5(2,2,2) Contact seepage; c_5(2,2,3) Bypass seepage; c_5(2,2,4) Diffuse seepage; c_5(2,2,5) Turbid seepage water
	c_5(2,3) Seepage stability problems	c_5(2,3,1) Insufficient seepage stability; c_5(2,3,2) Piping; c_5(2,3,3) Soil flow
	c_5(2,4) Abnormal seepage monitoring indicators	c_5(2,4,1) Abnormal changes in seepage discharge; c_5(2,4,2) Abnormal phreatic line; c_5(2,4,3) Abnormal uplift pressure
c_5(3) Structural Hazards	c_5(3,1) Structural design defects	c_5(3,1,1) Insufficient width; c_5(3,1,2) Insufficient elevation; c_5(3,1,3) Missing protective components
	c_5(3,2) Abnormal deformation and displacement	c_5(3,2,1) Excessive settlement; c_5(3,2,2) Abnormal displacement; c_5(3,2,3) Deformation failing to meet requirements
	c_5(3,3) Structural load-bearing and stability defects	c_5(3,3,1) Insufficient structural strength; c_5(3,3,2) Insufficient stability
c_5(4) Seismic Hazards	c_5(4,1) Seismic design and fortification defects	c_5(4,1,1) Insufficient seismic fortification intensity; c_5(4,1,2) Incorrect seismic fortification category; c_5(4,1,3) Incorrect design seismic parameters
	c_5(4,2) Seismic load-bearing and stability defects	c_5(4,2,1) Insufficient seismic strength; c_5(4,2,2) Insufficient seismic stability; c_5(4,2,3) Inadequate seismic measures; c_5(4,2,4) Potential structural liquefaction
c_5(5) Metal Structure Hazards	c_5(5,1) Layout and type-selection defects	c_5(5,1,1) Unreasonable layout; c_5(5,1,2) Unreasonable type selection; c_5(5,1,3) Insufficient lifting height of the hoisting mechanism; c_5(5,1,4) Insufficient span of the hoisting mechanism
	c_5(5,2) Service-life and durability defects	c_5(5,2,1) Exceeded the service life for depreciation and retirement; c_5(5,2,2) Surface corrosion and rusting; c_5(5,2,3) Seal failure
	c_5(5,3) Structural performance defects of metal structures	c_5(5,3,1) Insufficient strength of metal structures; c_5(5,3,2) Insufficient stability of metal structures
	c_5(5,4) Abnormal operational functions	c_5(5,4,1) Insufficient hoisting force; c_5(5,4,2) Jamming or failure in opening and closing operations; c_5(5,4,3) Abnormal safety protection devices; c_5(5,4,4) Abnormal locking or pressure-equalizing devices
	c_5(5,5) Abnormal auxiliary and support systems	c_5(5,5,1) Abnormal power supply; c_5(5,5,2) Abnormal control system

Table 4
Subclasses of "c_6 HazardComponent".

Subclass (level 1)	Subclass (level 2)	Subclass (level 3)
c_6(1) Water-retaining structures	c_6(1,1) Main dam structures	c_6(1,1,1) Dam crest; c_6(1,1,2) Dam abutment; c_6(1,1,3) Dam foundation; c_6(1,1,4) Dam toe; c_6(1,1,5) Dam body; c_6(1,1,6) Dam slope/surface; c_6(1,1,7) Wave wall; c_6(1,1,8) Flood control system
	c_6(1,2) Impervious body	c_6(1,2,1) Dam impervious body; c_6(1,2,2) Foundation impervious body
	c_6(1,3) Drainage system	c_6(1,3,1) Dam body drainage; c_6(1,3,2) Foundation drainage; c_6(1,3,3) Abutment drainage
c_6(2) Outlet structures	c_6(2,1) Spillway	c_6(2,1,1) Spillway intake/control section; c_6(2,1,2) Chute; c_6(2,1,3) Side wall; c_6(2,1,4) Spillway energy dissipation structures; c_6(2,1,5) Spillway metal structures
	c_6(2,2) Flood discharge tunnel	c_6(2,1,1) Spillway intake/control section; c_6(2,1,2) Chute; c_6(2,1,3) Side wall; c_6(2,1,4) Spillway energy dissipation structures; c_6(2,1,5) Spillway metal structures
	c_6(2,4) Other outlet structures	c_6(2,3,1) Bottom outlet; c_6(2,3,2) Emergency spillway; c_6(2,3,3) Emptying pipe; c_6(2,3,4) Other outlet metal structures
c_6(3) Conveyance structures	c_6(3,1) Intake structures	c_6(3,1,1) Intake tower; c_6(3,1,2) Inclined intake pipe; c_6(3,1,3) Metal structures of intake facilities
	c_6(3,3) Conveyance pipelines	c_6(3,2,1) Conveyance culvert; c_6(3,2,2) Conveyance tunnel; c_6(3,2,3) Metal structures of conveyance pipelines

3.2.2. *Rule-based knowledge extraction*

Entities associated with the top-level class c_7 represent safety evaluation results in dam safety reports, including flood-control, seepage, structural, seismic, and metal-structure safety evaluations. Each evaluation is classified into Grade A, B, or C, corresponding to no defect, minor defect, and serious defect, respectively. Because these results are generally described in highly standardized forms, they can be directly extracted using regular expressions. For example, from the expression "the flood-control safety rating is Grade C", the flood-control safety evaluation result can be accurately extracted as "C", without the need for deep learning-based recognition.

3.2.3. *LLM-based knowledge extraction*

In contrast, the extraction of engineering hazard entities is more challenging, with the core difficulty lying in how to accurately assign hazard information expressed in complex and diverse forms to the predefined category system of the DDO.

General-purpose language models achieve relatively low accuracy on this highly specialized task, as also supported by the experimental results presented in Section 4.2.3. Although general-purpose LLMs show limited performance in fine-grained classification, under explicit instruction constraints they are still able to extract hazard-related descriptive fragments from engineering reports efficiently, that is, they can effectively determine which parts of the text correspond to hazard

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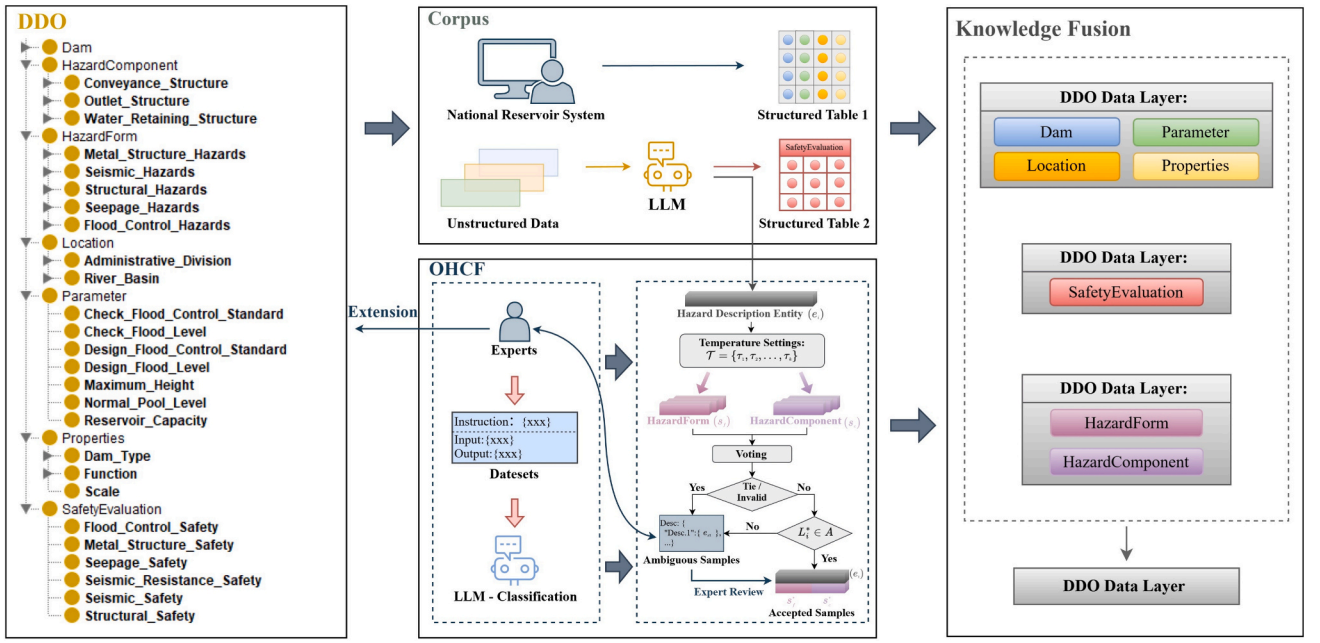


Fig. 3. Framework for data layer construction.

descriptions. Accordingly, given an input report text T, the extraction prompt P_1 is first used to identify hazard-related descriptive fragments. However, the results obtained at this stage are usually still descriptive fragments directly extracted from the original text, and often contain nested expressions, meaning that a single sentence may simultaneously include multiple combinations of hazard forms and hazard components. This creates an obstacle to subsequent hazard classification. To address this issue, this study further designs a description-splitting prompt P_2 to decompose complex hazard expressions into hazard description units that are semantically more explicit and structurally more singular, as shown in Fig. 4. The above two-step process can be formulated as Equation (3).

$$T \xrightarrow{P_1} H = \{h_1, h_2, \ldots, h_m\} \xrightarrow{P_2} E = \{e_1, e_2, \ldots, e_n\} \tag{3}$$

where h_i denotes a hazard-related descriptive fragment extracted from the original text, and e_i denotes a hazard description unit obtained after description splitting. Based on our instance-level observations, general-purpose LLMs can perform both tasks in a relatively stable manner using prompts alone, without task-specific training.

On this basis, OHCF is designed to classify each hazard description unit into the third-level labels of classes c_5 and c_6 defined in the DDO. OHCF consists of four stage:

a) LLM fine-tuning.

To improve the model's ability to perform fine-grained classification under limited computational resources, low-rank adaptation (LoRA) is

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===== PROMPT =====
You are an expert in processing textual descriptions from dam safety appraisal reports.
Please semantically decompose the input hazard description into several hazard description units with clear and singular meanings.

Requirements:
1. Each output sentence should contain only one hazard component and one hazard form.
2. If multiple hazard forms share the same hazard component in the original text,
   the shared component should be explicitly completed in each decomposed sentence.
3. Do not change the original meaning or add information that is not present in the input text.
4. List the results as complete hazard description sentences.

===== INPUT =====
The downstream slope of the dam lacks slope protection, and there are issues
with an elevated phreatic line exit point and multiple areas of diffuse seepage.

===== MODEL =====
deepseek-v3

===== OUTPUT =====
1. The downstream slope of the dam lacks slope protection.
2. The downstream slope of the dam has an elevated phreatic line exit point.
3. The downstream slope of the dam has multiple areas of diffuse seepage.
```

Fig. 4. Example of nested hazard description splitting using prompting.

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adopted to fine-tune a lightweight LLM, enabling it to learn the semantic mapping between hazard description units and the ontology-defined labels.

LoRA is an efficient parameter-adaptation method for fine-tuning large models [57]. It requires updates to only a small number of parameters to improve performance on downstream tasks. The core principle of LoRA is to freeze the original weight matrix $W_0 \in \mathbb{R}^{m\times n}$ of the model and update only the parameters of the incremental matrix ΔW. The incremental update can be expressed as Equation (4), where ΔW is decomposed into two smaller matrices A and B. The output h of the fine-tuned model is obtained by summing the products of the original weight matrix W_0 and the incremental matrix ΔW with the same input vector x, as shown in Equation (5).

$$W_0 + \Delta W = W_0 + BA, A \in \mathbb{R}^{(r\times n)}, B \in \mathbb{R}^{(m\times r)} \tag{4}$$

$$h = W_0 x + \Delta W x = W_0 x + BAx \tag{5}$$

After fine-tuning, the functionality of the model can be formally described as a function f_{θ^*}, which generates corresponding standardized entity labels for any candidate entity:

$$f_{\theta^*} : e_i \rightarrow (s_f, s_c), s_f = c_5(j_1, j_2, j_3), s_c = c_6(k_1, k_2, k_3) \tag{6}$$

b) Temperature-perturbed voting strategy for hazard classification

For generative models, the result of a single inference is not necessarily the globally optimal solution. To further reduce the probability of misclassification, a voting mechanism is introduced. Specifically, n different model temperatures $\mathcal{T} = \{\tau_1, \tau_2, ..., \tau_n\}$ are used to independently perform n inference runs for a given hazard description entity e_i. The frequency with which each candidate label appears across these runs is then counted as its number of votes. Following the majority principle, the candidate labels that receive the highest number of votes are selected as the final and most reliable classification result L_i^*, as defined in Equations (7) – (9).

$$L_i^* = (s_f^*, s_c^*) \tag{7}$$

$$s_f^* = \underset{s_f \in S_5}{\operatorname{argmax}} \sum_{k=1}^{n} \mathbb{I}(s_{f,k} = s_f) \tag{8}$$

$$s_c^* = \underset{s_c \in S_6}{\operatorname{argmax}} \sum_{k=1}^{n} \mathbb{I}(s_{c,k} = s_c) \tag{9}$$

where $s_{f,k}$ and $s_{c,k}$ are the c_5 class and c_6 class obtained from the $k-$th independent inference of e_i, respectively, while s_f^* and s_c^* are the highest-scoring results among the k inference results, that is, the optimal solutions.

For samples where the model fails to produce classification labels or where the voting results in a tie, the system uniformly assigns a "*False*" identifier and forwards these samples to Stage d for further processing.

c) Ontology-constraint consistency checking.

Although the temperature-perturbed voting strategy improves the stability of hazard classification, the model may still generate clearly inconsistent label pairs. For example, if the hazard description "the existing wave wall crest elevation is insufficient" is classified after voting as the label pair (s_f^*: c_5(1,2,1) Insufficient water-retaining elevation, s_c^*: c_6(1,2,1) Dam impervious body), the resulting pair is obviously inconsistent with engineering knowledge.

To further improve the reliability of classification results, this study introduces an ontology-constraint consistency checking module. A constraint-conflict set A, defined as a sparse subset of the cartesian product, is introduced to record clearly incompatible c_5 and c_6 label pairs according to the DDO hierarchy and engineering knowledge, as expressed in Equation (10).

$$A \subseteq S_f \times S_c \tag{10}$$

Based on this constraint-conflict set, a classification consistency checking function V is defined. V takes the final classification result $L_i^* = (s_f^*, s_c^*)$ from Stage b as input, determines whether the classification result violates the predefined ontology constraints, and outputs a Boolean value:

$$V(L_i^*) = V(s_f^*, s_c^*) = \begin{cases} False & \text{if } (s_f^*, s_c^*) \in A \\ True & \text{if } (s_f^*, s_c^*) \notin A \end{cases} \tag{11}$$

Entity samples for which $V(L_i^*) = True$ are accepted, but samples with $V(L_i^*) = False$ are forwarded to Stage d for further handling.

d) Expert review and ontology expansion

Samples marked as "*False*" are automatically aggregated by the system and submitted to an expert review queue, where they await manual evaluation by domain experts.

During the review process, experts first determine whether the concept corresponding to a given sample already exists in the DDO. If the concept is already present, experts correct the classification label by linking the hazard description unit to the correct existing ontology node. If the expert confirms that the concept is new, a corresponding node is added to the DDO, and the hazard description entity is then associated with this newly created node. In this semi-automated workflow, the system automatically filters ambiguous samples and experts perform final label correction and ontology evolution. In this way, the DDO achieves dynamic updating and iterative expansion.

3.3. KG storage

In this paper, the Neo4j graph database is selected as the underlying storage platform for DDKG. Utilizing the Cypher query language, Neo4j enables efficient knowledge querying, addition, deletion, and modification. It also exhibits superior performance in executing deep associative queries and multi-hop reasoning [58], providing robust support for subsequent knowledge retrieval and interaction.

The knowledge extracted in Section 3.2 is integrated into JSON format. Based on the semantic and hierarchical relationships within the DDO, a Python script was developed to parse JSON key-value pairs and generate corresponding Cypher statements. This process facilitates the batch creation of DDKG nodes and the construction of inter-entity relationships, completing the injection of dam hazard knowledge into the Neo4j database.

4. Case study

This section begins by outlining the data sources for the case study. Subsequently, ablation experiments are conducted to systematically validate the effectiveness of the key components of OHCF. Finally, the DDKG is established, and its application in dam defect analysis is demonstrated.

4.1. Data sources

In China, dam projects are categorized into Class I, II, and III based on their safety status. Class III dams, which are characterized by significant safety hazards, offer safety appraisal reports that serve as an invaluable data source for risk information. The knowledge in the DDKG is derived from two primary datasets: (1) basic engineering attributes, hydrological records, and geospatial data collected from the China Reservoir Operation and Management Platform; and (2) safety appraisal reports of 22,076 distressed dams across China recorded between 2000 and 2025.

The safety assessment reports of dams in China have a unified standardized structure. The entities related to the top-level classes c_5 to c_7 exist in the following four parts of the report: on-site inspection (including engineering appearance inspection, geological exploration,

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etc.), dam safety analysis and evaluation (sub-item analysis on aspects such as flood control, seepage, and structure of the project), summary of main issues, and safety assessment conclusions. Using the preprocessing method in Section 3.2.1, these four main sections in each dam safety assessment report are identified and extracted through regular expressions, serving as the corpus for subsequent entity recognition.

4.2. Model training and evaluation

4.2.1. Dataset construction

Currently, there are no publicly available model training datasets for hydraulic engineering safety hazards. Therefore, this study constructs a domain-specific dataset using a semi-automated annotation workflow, which consists of three stages: preliminary machine labeling [59], expert verification, and data augmentation [60]. DeepSeek-V3-671B was first used to generate preliminary labels for the extracted hazard description samples. These labels were then verified and corrected by domain experts, and data augmentation was subsequently performed based on the expert-corrected samples.

Fig. 5 illustrates the comprehensive workflow for the fine-tuning dataset construction. The dataset structure adopts the Alpaca format [61], which follows the "Instruction-Input-Output" triplet paradigm. In this framework, "Input" consists of pre-processed raw report text, while "Output" represents the corrected structured entity set. The "Instruction" is composed of three elements: role setting, task description, and output requirements. Role setting is designed to assign a specific identity and function to the model, guiding its comprehension of the task background; the task description defines the specific operations and objectives the model is expected to execute; and the output requirements dictate the desired format and content of the results—in this study, the output format is specified as JSON format.

A total of 120,471 hazard description samples were extracted and subjected to initial machine labeling. These annotated samples were mapped to the five hazard categories through the hierarchical relationships defined in the DDO. Subsequently, 1,200 samples were randomly selected from the annotated corpus for manual verification and validation by experts. To further enhance the dataset, the LLM) was prompted to perform semantic rewriting on the original hazard descriptions. This process generated new instances that are semantically equivalent but linguistically diverse, thereby enriching the training data. The statistics for the hazard samples at each stage of the process are summarized in Table 5.

Table 5
Number of hazard samples at each stage.

Type	Sample size	Expert-corrected samples	Data augmentation
Flood control	210	241	1205
Seepage	313	339	1695
Structural	372	353	1765
Seismic	187	188	940
Metal structure	218	179	895
Total	1300	1300	6500

4.2.2. Experimental configuration and evaluation metrics

Given that the tasks in this study are based on Chinese engineering texts, we selected open-source LLMs trained primarily on Chinese corpora for the experiments, including DeepSeek-V3-671B, Qwen3-32B, GLM4-32B, Qwen3-14B, GLM4-9B, and Qwen3-8B. All experiments were conducted on a server equipped with eight NVIDIA H800 80 GB GPUs, using Python 3.10 and CUDA 12.6. Model fine-tuning was implemented with the LLaMA-Factory framework, and no model quantization was applied. It should be noted that DeepSeek-V3-671B was not fine-tuned due to limitations in computational resources and dataset size; instead, it was used only as a baseline reference.

During fine-tuning, the maximum learning rate was set to 5×10^{-4}, and the number of training epochs was set to 7. The dataset was divided into training, validation, and test subsets at a ratio of 7:1:2. To reduce the risk of overfitting, an early stopping strategy was adopted [62]. Specifically, fine-tuning was terminated in advance if the validation loss continued to increase over several epochs.

Specifically, fine-tuning was terminated in advance if the validation loss continued to increase over several epochs "False" were regarded as null-output cases, indicating that no target entity was identified. The evaluation metrics included macro-averaged precision P_{macro}, macro-averaged recall R_{macro}, and macro-averaged F1-score $F1_{\text{macro}}$, with the calculation formulas presented in Equations (12) to (14).

$$P_{macro} = \frac{1}{n}\sum_{i=1}^{n} P_i = \frac{1}{n}\sum_{i=1}^{n} \frac{TP_i}{TP_i + FP_i} \tag{12}$$

$$R_{macro} = \frac{1}{n}\sum_{i=1}^{n} R_i = \frac{1}{n}\sum_{i=1}^{n} \frac{TP_i}{TP_i + FN_i} \tag{13}$$

$$F1_{\text{macro}} = \frac{1}{n}\sum_{i=1}^{n} F1_i = \frac{1}{n}\sum_{i=1}^{n} 2 \cdot \frac{P_i \cdot R_i}{P_i + R_i} \tag{14}$$

where TP_i is the number of samples whose true labels are class i and are

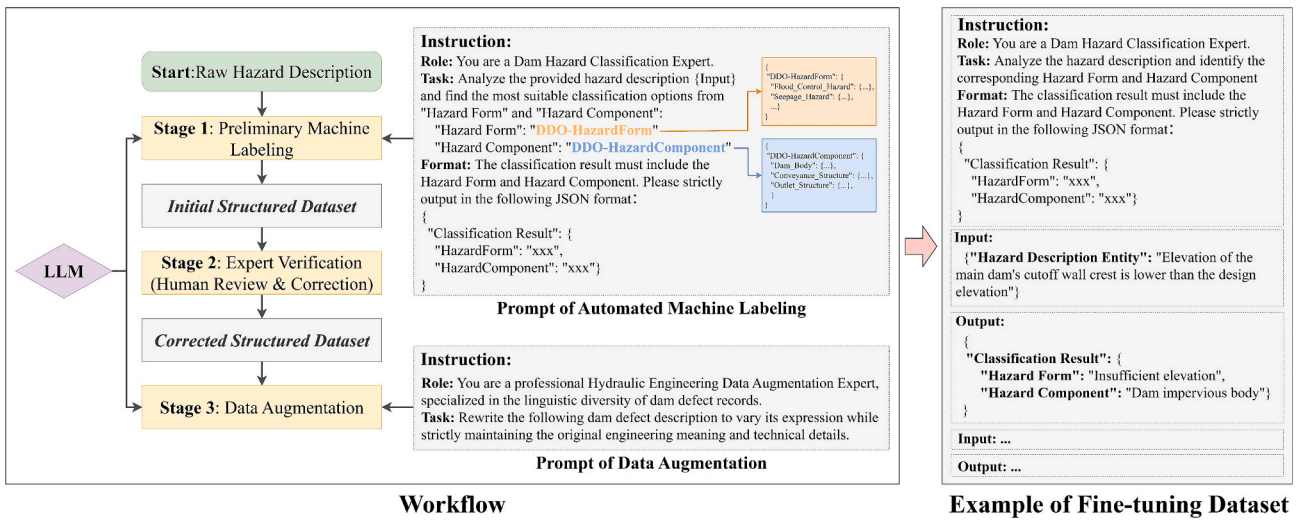


Fig. 5. Dataset construction.

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correctly predicted by the model, FP_i is the number of samples whose true labels belong to other classes but are incorrectly predicted as class i, and FN_i is the number of samples whose true labels are class i but are incorrectly predicted as other classes.

4.2.3. Experimental evaluation

(1) Fine-tuning performance evaluation

The fine-tuning results are shown in Fig. 6. To ensure comparability across different models and between the pre- and post-fine-tuning results, all models were evaluated using the same prompt template during inference, in which the complete set of candidate labels was provided. The temperature parameter was uniformly set to 0 to reduce the influence of random sampling on model outputs. From the baseline results, the overall recognition performance generally improves as the model parameter scale increases. DeepSeek-V3-671B achieves the best baseline performance in both class c_5 and class c_6. This indicates that ultra-large general-purpose LLMs have stronger basic capabilities in semantic understanding of dam hazard descriptions and fine-grained category discrimination without domain-specific fine-tuning. This result also supports the use of large-parameter models in the data pre-annotation stage of this study, as they can generate relatively high-quality candidate labels and thereby reduce the workload required for subsequent manual correction.

After fine-tuning, the fine-tuned models showed substantial improvements in both class c_5 and class c_6 recognition tasks, with $F1_{macro}$ increasing by approximately 26–34 percentage points compared with the baseline results. This indicates that domain-specific fine-tuning effectively improves the model's capability for hazard classification. Among the tested models, GLM4-32B achieved the best overall performance, with all three evaluation metrics approaching 90 % for class c_5 and also close to 90 % for class c_6. Qwen3-32B also achieved strong results after fine-tuning, although its metrics were slightly lower than those of GLM4-32B. In comparison, the 8B–14B models showed lower overall performance than the 32B models, but they require fewer computational resources and lower deployment costs, while still achieving relatively stable recognition results after fine-tuning. Therefore, in resource-constrained or cost-sensitive scenarios, 8B–14B models remain a practical option that balances performance and efficiency. For the subsequent experiments in this study, GLM4-32B was selected as the base model to ensure higher accuracy and stability in dam hazard entity recognition.

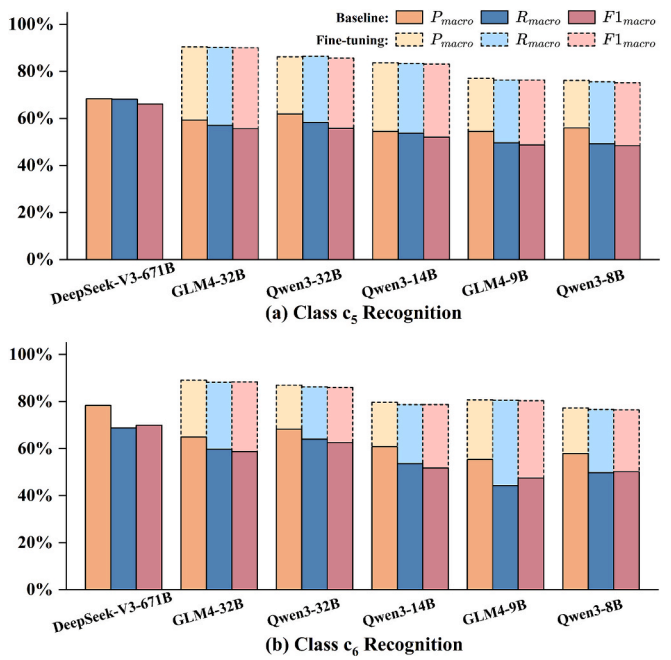


Fig. 6. Model performance before and after fine-tuning.

Further inspection of the prediction results shows that the remaining errors in class c_5 recognition are mainly associated with structural hazards. For structural hazards, the models tend to confuse fine-grained categories such as insufficient structural strength and structural instability. This is mainly because these categories are described in highly similar ways in engineering reports. Expressions such as "not satisfying specification requirements", "insufficient safety", and "inadequate bearing capacity" are frequently used, but the text does not always clearly indicate whether the problem refers to material- or member-level bearing capacity, or to the overall stability, sliding stability, or overturning stability of the structure. As a result, the models can generally identify such descriptions as structural safety problems, but still face difficulties in aligning them with the correct fine-grained hazard form. In contrast, the recognition results for class c_6 are relatively balanced across different categories, with no obvious performance bottleneck. This indicates that the models can identify dam components.

(2) Performance evaluation under different temperature parameters

Furthermore, we conducted voting experiments under temperature perturbations using the fine-tuned GLM4-32B and Qwen3-32B models. For the entity classification task, adjusting the temperature can influence output diversity. However, excessively high values may lead to overly divergent outputs and the generation of irrelevant content. In our experiments, six temperature values between 0 and 1 were tested, specifically $\mathscr{T} = \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$. The resulting performance variations of the models across these temperature settings are presented in Fig. 7.

The experimental results show that model performance under a single temperature setting fluctuates with changes in temperature, and the trends vary across models and recognition tasks. GLM4-32B maintains high overall performance in both the c5 and c6 tasks, but noticeable fluctuations occur at certain temperature settings, such as 0.4 and 0.6. In contrast, Qwen3-32B shows a slight overall decline in task performance as the temperature increases, suggesting that higher temperatures may affect output stability to some extent. Overall, the P, R, and F1 values obtained through multi-temperature voting are higher than those obtained under single-temperature settings across all experimental groups. This indicates that the voting strategy can effectively integrate model outputs under different temperature conditions and reduce the influence of single-inference errors on the final classification results, thereby improving the stability and overall performance of entity recognition.

In addition, the experiment further examined a cross-model hybrid voting strategy. Specifically, the predictions of Qwen3-32B and GLM4-32B under their relatively optimal temperature settings were selected, and the final labels were generated through majority voting. The results show that the overall performance of hybrid voting did not exceed that of the better-performing GLM4-32B model alone. A possible reason is that the overall prediction quality of Qwen3-32B was relatively lower, and its incorrect predictions may have weakened the decision advantage of GLM4-32B during voting. In addition, the two models may have shared similar confusion patterns for some fine-grained categories, preventing cross-model voting from forming effective complementarity.

(3) Ablation study.

The ablation study was conducted based on the GLM4-32B model to evaluate the contribution of the main components of OHCF. As shown in Table 6, fine-tuning provides the dominant performance gain, demonstrating its importance for adapting the model to the dam hazard classification task. The voting and consistency checking modules bring further but relatively modest improvements. Overall, the complete OHCF achieves the best performance for both c_5 and c_6 classification, with all evaluation metrics exceeding 90 %.

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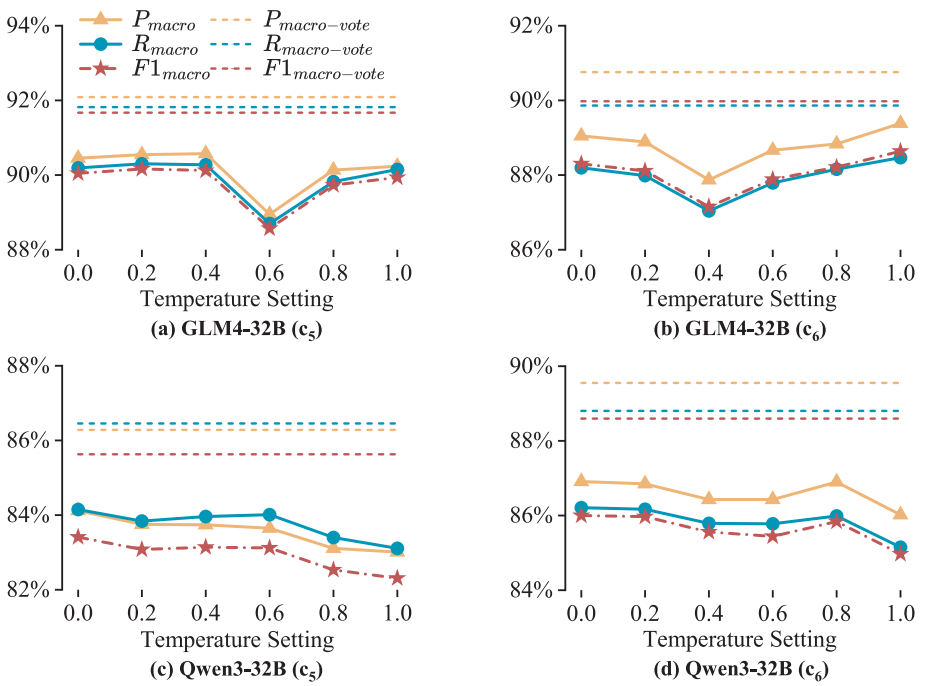


Fig. 7. Model performance under temperature perturbation and voting strategy.

Table 6
Ablation results of OHCF based on GLM4-32B.

	Class c_5 Classification			Class c_6 Classification		
	P_{macro}	R_{macro}	$F1_{macro}$	P_{macro}	R_{macro}	$F1_{macro}$
Baseline	59.29 %	57.06 %	55.65 %	64.88 %	59.69 %	58.66 %
w/o Voting & Consistency checking	90.45 %	90.19 %	90.04 %	89.05 %	88.20 %	88.30 %
w/o Consistency Checking	92.09 %	91.82 %	91.67 %	90.76 %	89.86 %	89.98 %
OHCF	93.26 %	93.01 %	92.87 %	91.55 %	90.62 %	90.76 %

4.3. Construction and retrieval application of DDKG

4.3.1. DDKG construction and visualization

Based on the experimental results, the fine-tuned GLM4-32B model was adopted as the backbone of OHCF and applied to all hazard description texts. The data layer entities associated with each engineering instance were organized into independent JSON files and batch-imported into the Neo4j graph database using Python scripts. In total, the constructed DDKG contained 201,362 nodes and 953,447 relationships. The number of nodes associated with each top-level class is listed in Table 7. Parameter nodes account for a relatively large proportion because each dam instance is described by multiple engineering and hydrological indicators, including dam height, reservoir capacity, and characteristic water levels.

Fig. 8 illustrates the main knowledge structure of the DDKG. Each dam instance is semantically mapped to its corresponding entity node. Under the semantic constraints defined in the dam domain ontology (DDO), each dam engineering entity is connected to its associated attribute, evaluation, hazard-form, and hazard-component entities. These entities can be further traced upward along the ontology hierarchy to establish multi-level semantic associations with their parent nodes (e.g., Dam → c_5(5,2,2) ← c_5(5,2) ← c_5(5)). In addition, engineering entities can be topologically connected across different projects through shared entity nodes. For example, two dam instances in Fig. 8 are associated through common nodes such as “River Basin: Pearl River Basin”, “Scale: Small”, and “Dam type: Homogeneous Earth Dam”. This interconnected structure provides a robust foundation for advanced tasks such as clustering analysis of dam hazard knowledge and knowledge reasoning.

Table 7
Number of nodes associated with each top-level class.

c_i	Entity class	Quantity
c_1	Dam	22,076
c_2	Properties	58
c_3	Location	1,922
c_4	Parameter	76,152
c_5	HazardForm	73
c_6	HazardComponent	44
c_7	SafetyEvaluation	15
Total		100,340

4.3.2. Example of multi-constraint retrieval

The core function of KGs lies in efficient knowledge retrieval. By constraining key entity nodes, the DDKG can retrieve the distribution of engineering hazards under specific boundary conditions. Fig. 9 illustrates the retrieval of structural safety hazards under multiple constraints. This retrieval involves three fixed constraints: Yangtze River Basin (River_Basin), embankment dam (Dam_Type), and medium (Scale). These constraints restrict the search scope to all dam projects corresponding to this combination. Within the defined sample pool, hazard nodes associated with these projects are traced via the (:Dam)-[:exhibitsHazards]->(:HazardForm) relationship. At the same time, constraints on hazard forms are constructed based on the hierarchical structure under the top-level class c_5:HazardForm in the DDO: (:HazardForm {node_name:'Structure'})-[:hasSubclass*0..]->(: HazardForm). This strictly limits the retrieval to the “Structure Hazards” node and all

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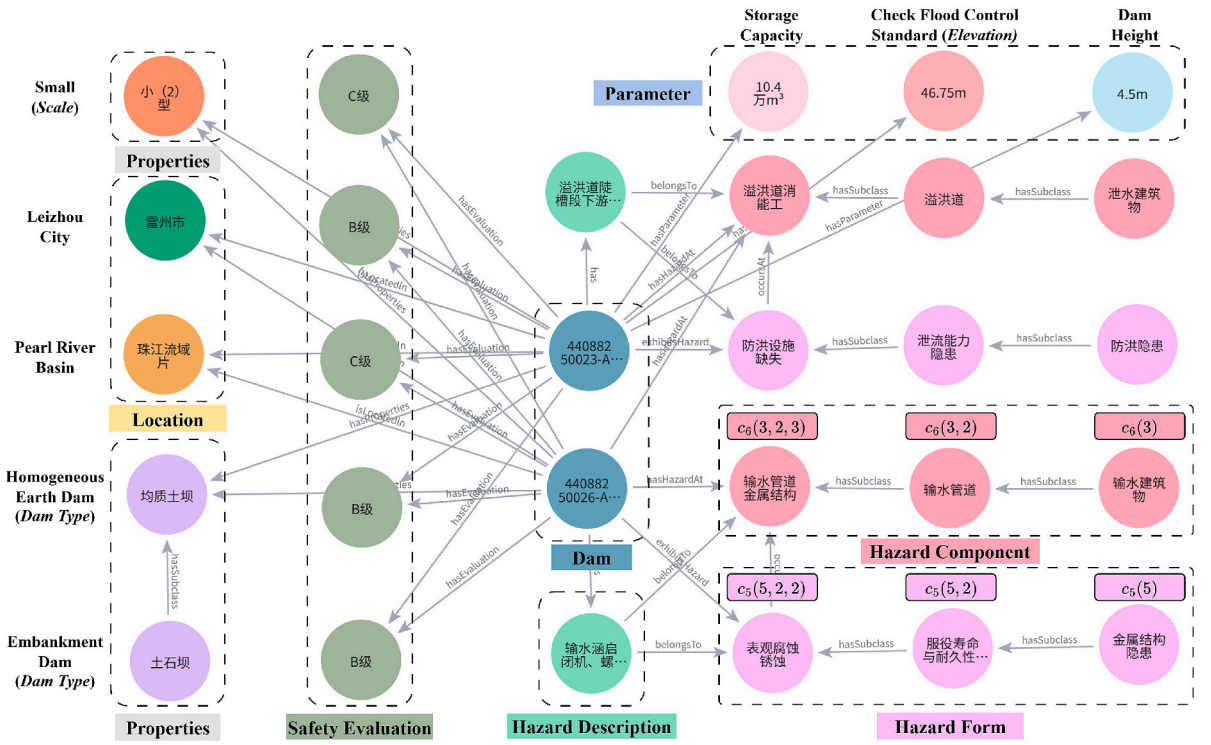


Fig. 8. Visualization of the main DDKG structure.

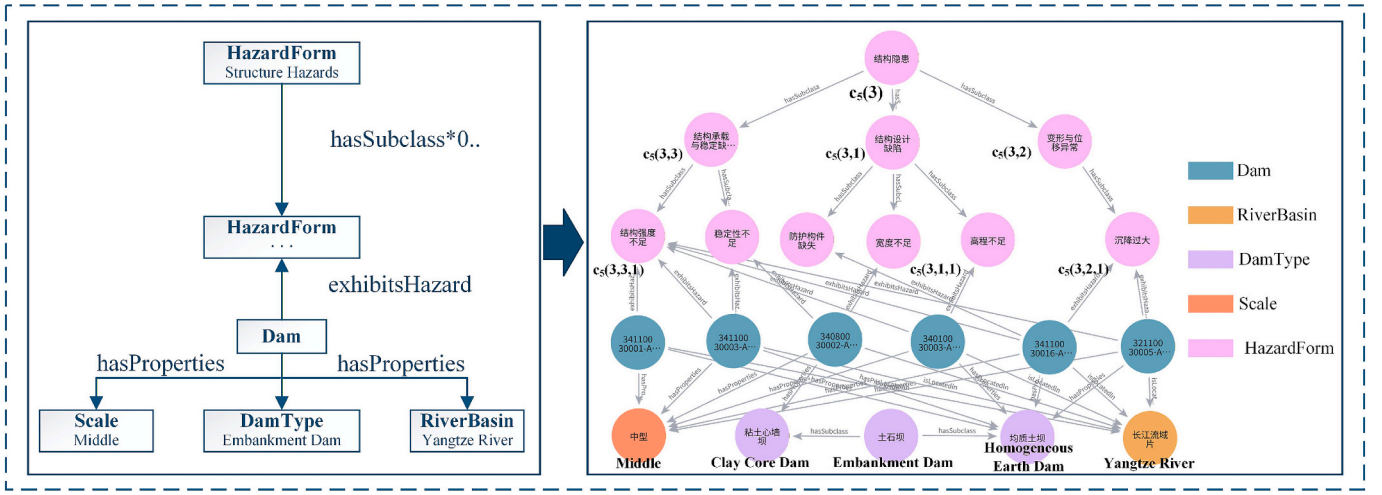


Fig. 9. Example of Structural Hidden Danger Knowledge Retrieval.

its recursive subclasses.

The DDKG, as a structured knowledge base, organizes originally scattered information on river basins, dam types, and defective and dangerous dam hazards into an ordered network through entity and relationship links. In this way, complex retrieval and reasoning tasks—such as multi-conditional filtering or cross-node risk tracing—can be performed based on graph paths, rather than simple text matching.

4.4. DDKG-based analysis of dam defects in the Yangtze River Basin

This section takes dam projects in the Yangtze River Basin as the research object and conducts a case analysis of dam safety defects. The purpose is to further verify the application potential and value of the DDKG in engineering defect analysis and decision-support tasks.

In practical engineering, a Grade-C result in a specific safety evaluation category indicates that the project has relatively serious safety defects in that category. To identify the major safety weaknesses of dam projects in the Yangtze River Basin, the occurrence frequencies and co-occurrence frequencies of different Grade-C safety evaluation indicators were statistically analyzed, as shown in Fig. 10. In the figure, $c_7(1)$, $c_7(2)$, $c_7(3)$, $c_7(4)$, and $c_7(5)$ represent flood-control safety, seepage safety, structural safety, seismic safety, and metal-structure safety evaluations, respectively. The diagonal values indicate the occurrence frequency of each Grade-C safety evaluation indicator.

From the perspective of individual safety evaluation results, Grade-C problems in dam projects of the Yangtze River Basin are mainly concentrated in seepage safety, structural safety, and flood-control safety. Among them, seepage safety has the highest Grade-C occurrence frequency, indicating that seepage-related defects are the most

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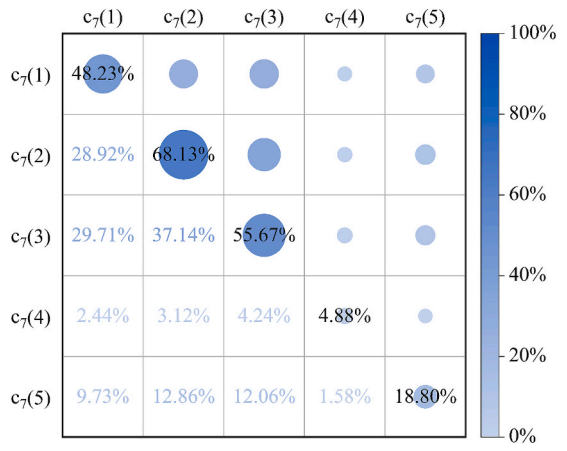


Fig. 10. Occurrence and co-occurrence frequencies of Grade-C safety evaluation indicators in the Yangtze River Basin.

prominent weakness in the safety evaluation of dam projects in this basin. The Grade-C occurrence frequencies of structural safety and flood-control safety are also relatively high, reaching 55.67 % and 48.23 %, respectively. This suggests that dams in the Yangtze River Basin commonly face problems such as structural aging, insufficient seepage-control capacity, and inadequate flood-control capacity during long-term operation. In contrast, metal-structure safety defects and seismic defects are not the dominant problems for dam projects in the Yangtze River Basin.

4.4.1. Analysis of seepage–structure compound defects

The co-occurrence analysis of Grade-C safety evaluation indicators shows a relatively strong association between seepage defects and structural defects in the Yangtze River Basin. To further identify the specific defect combinations underlying this association, the top five co-occurrence patterns of severe seepage and structural defects were extracted, as shown in Table 8.

It can be seen that seepage leakage "c_5(2,2,1)" in conveyance culverts "c_6(3,2,1)" has the highest co-occurrence probability with insufficient structural strength "c_5(3,3,1)". This combination represents the most typical seepage–structure compound defect in the Yangtze River Basin. In early earth-rockfill dams, conveyance culverts were mostly embedded within the dam body and were commonly built as masonry structures. Their material properties differ markedly from those of the surrounding dam-fill materials, making it difficult to maintain coordinated seepage deformation and stress deformation. As a result, concentrated seepage paths are likely to develop along the contact zones between the culvert and the dam body. Long-term seepage erosion may not only weaken the seepage stability of these contact zones, but also accelerate the deterioration of the masonry culvert structure. This may reduce structural strength and further induce or aggravate culvert leakage.

In addition, seepage leakage c5(2,2,1) in dam foundations c6(1,1,3) is also likely to be coupled with structural instability defects. Long-term foundation seepage may cause soil loss or local erosion at the contact zones between the dam foundation and the dam body, thereby weakening the supporting conditions of the dam. This process may induce excessive settlement c5(3,2,1) of dam slopes c6(1,1,6), and may even develop into overall sliding instability c5(3,3,2). Meanwhile, uneven settlement of the dam body or foundation may also adversely affect auxiliary structures such as the spillway c6(2,1,1), leading to structural cracking or deformation at contact zones, which may further induce seepage problems.

Therefore, seepage anomalies may weaken the stability conditions of the dam body and foundation, while long-term seepage action may further accelerate structural deterioration and local damage. Conversely, structural defects can also aggravate seepage problems by inducing cracking, deformation, or contact-zone failure, and may further affect the overall flood-control capacity and operational safety of the project. These results indicate that seepage and structural defects in the Yangtze River Basin are not independent problems, but are often coupled and mutually reinforcing. Therefore, safety diagnosis and reinforcement measures should focus on the combined effects of seepage control, structural stability, and component integrity, rather than treating seepage and structural defects as isolated issues.

Table 8
Top five co-occurrence combinations of seepage and structural defects.

No.	Seepage Hazards		Structure Hazards		Co-occurrence Frequency
	Class c_5	Class c_6	Class c_5	Class c_6	
1	c_5(2,2,1)	c_6(3,2,1)	c_5(3,3,1)	c_6(3,2,1)	13.99 %
2	c_5(2,2,1)	c_6(1,1,3)	c_5(3,3,2)	c_6(1,1,6)	13.83 %
3	c_5(2,2,1)	c_6(1,1,3)	c_6(1,1,6)	c_6(2,1,1)	13.55 %
4	c_5(2,2,1)	c_6(1,1,3)	c_5(3,2,1)	c_6(1,1,6)	12.96 %
5	c_5(2,2,1)	c_6(1,1,3)	c_5(3,3,1)	c_6(3,2,1)	10.57 %

4.4.2. Flood control hazard analysis of embankment dams

Flood control and disaster mitigation are among the primary functions of reservoir dams. For embankment dams, insufficient flood-control capacity may directly increase the risk of overtopping and threaten the overall safety of the dam.

Fig. 11 depicts the frequency distribution of various flood control hazard forms for embankment dams of different project scales in the Yangtze River Basin. Panels (a), (b), and (c) correspond to small, medium, and large projects, respectively. The occurrence frequency of each flood control hazard was calculated using Equation (15):

$$P\left(c_5(1,j_2,j_3)|\lambda_k,\Omega\right)=\frac{N_{dam}\left(c_5(1,j_2,j_3)\cap\lambda_k\cap\Omega\right)}{N_{dam}(\lambda_k\cap\Omega)}\times 100\% \tag{15}$$

where λ_k is the project scale, with k = 1, 2, and 3 corresponding to small, medium, and large projects, respectively; Ω is the constraint set ({Yangtze River Basin (River_Basin), embankment dam (Dam_Type)}; $N_{dam}(\lambda_k\cap\Omega)$ is the number of dams with "project scale = λ_k & Ω"; and $N_{dam}\left(c_5(1,j_2,j_3)\cap\lambda_k\cap\Omega\right)$ is the number of dams with "flood control hazard exists $c_5(1,j_2,j_3)$ & project scale = λ_k & Ω."

Overall, "c_5(1,2) Insufficient freeboard of water-retaining structures" and "c_5(1,3) Discharge capacity hazards" emerge as the most critical flood control hazards. Within these categories, "c_5(1,2,1) Insufficient dam crest elevation" and "c_5(1,3,2) Insufficient discharge capacity of the spillway" are the most frequently occurring specific hazard forms.

Small-scale projects are more prone than medium- and large-scale projects to issues such as "c_5(1,1,2) Insufficient design flood standard" and "c_5(1,2,1) Insufficient dam crest elevation." This reflects the limitations of small reservoirs, which—constrained by lower fortification standards and limited technical investment during earlier construction periods— not only struggle to meet current design codes in terms of technical parameters but also lack sufficient structural height margins to safely accommodate extreme conditions.

The occurrence frequency of "c_5(1,3,2) Insufficient discharge capacity" increases with project scale. Compared with small-scale projects, medium- and large-scale projects generally face greater flood pressures and more complex flood routing processes. These characteristics place higher demands on the discharge capacity of hydraulic structures, the efficiency of operational response, and the coordination of the overall flood release system. As a result, mismatches between discharge capacity and actual flood control demands are more likely to occur in larger projects.

In small projects, although insufficient discharge capacity is relatively less prominent, "c_5(1,3,1) Missing flood control facilities" is more severe. This is mainly because some small reservoirs have remained empty for many years owing to prolonged periods without inflow. Under

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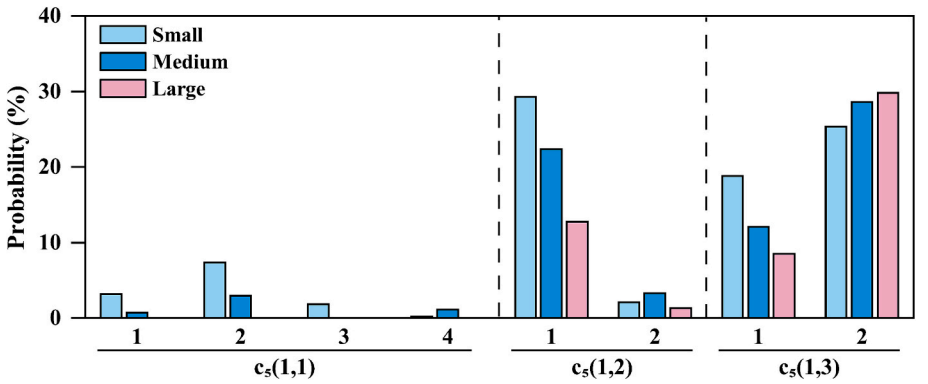


Fig. 11. Frequency distribution of flood control hazard forms.

such conditions, their original water release or conveyance facilities may gradually fall into disuse and become inoperative, leading to a substantial weakening of flood control function. Although these reservoirs may currently have little practical need for discharge, the associated risk should not be overlooked. Once a flood event occurs, the reservoir may rapidly shift from an empty state to an impounding state. In the absence of functional flood release facilities, inflow cannot be discharged in time, which greatly increases the risk of overtopping and, in severe cases, dam failure.

In summary, the DDKG provides a structured basis for multi-dimensional dam hazard analysis. The Yangtze River Basin case study shows that hazard information can be retrieved and compared by basin, dam type, project scale, hazard type, and affected component. This demonstrates the potential of the DDKG to support regional dam safety analysis and management decision-making.

5. Conclusion and future work

To address the underutilization of unstructured text data in the dam safety domain and the challenges of knowledge adaptation when applying general-purpose LLMs to specialized fields, this study proposes a KG construction method that integrates a domain knowledge ontology with LLMs. The main contributions of this research are as follows:

(1) A dam defects ontology (DDO) was developed to provide a structured representation framework for dam hazard knowledge. In particular, the classes of hazard forms and affected components were refined to support fine-grained modeling of dam defects.

(2) An Ontology-guided Hazard Classification Framework (OHCF) was proposed for standardizing unstructured hazard descriptions. By integrating fine-tuning, temperature-perturbed voting, ontology-constraint consistency checking, and expert validation, OHCF improves the reliability of hazard classification and provides structured inputs for KG construction.

(3) Based on the proposed method, a dam defects knowledge graph (DDKG) was constructed using safety appraisal reports from 22,076 dams. The case study shows that the DDKG can support multi-constraint retrieval and regional hazard analysis, demonstrating its potential value for dam safety management.

Despite these contributions, the proposed method has limitations that will be addressed in future work:

(1) The application of the DDKG in this study remains preliminary, focusing mainly on a single large-scale regional analysis. Future research will further expand its application to more refined spatial-scale and project-level hazard analyses, to reveal regional differences in hazard distribution and provide more targeted support for dam safety management.

(2) At present, the DDKG is a static knowledge base, as its hazard information is derived from historical evaluation reports. Future work will incorporate temporal data and explore automated update methods to transform the DDKG into a dynamic, evolving system.

(3) Although this study evaluates hazard knowledge extraction and includes expert validation, the constructed knowledge graph itself still requires systematic assessment. Future research will examine its structural properties, completeness, consistency, and practical effectiveness in supporting dam defect analysis and engineering decision-making.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Gemini 2.5 in order to improve the language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Jianfeng Huang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Wenang Hou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation. **Shichen Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Zhenhan Du:** Visualization, Investigation, Data curation. **Jiangshan Yin:** Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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